

EX NO:	Performance Analysis of IIR Filters
DATE:	3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB.

PROGRAM

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clc; clear; close all;
% Specifications Fs=2000; Rp=1; Rs=40; Wp=[300 600]/(Fs/2); Ws=[200 700]/(Fs/2);

% Butterworth Filter [n1,W1]=buttord(Wp,Ws,Rp,Rs);
[b1,a1]=butter(n1,W1,'bandpass');

% Chebyshev Type-I Filter [n2,W2]=cheb1ord(Wp,Ws,Rp,Rs);
[b2,a2]=cheby1(n2,Rp,W2,'bandpass');

% Magnitude Response figure;
freqz(b1,a1); title('Butterworth Filter');

figure;
freqz(b2,a2); title('Chebyshev Type-I Filter');

% Group Delay figure;
grpdelay(b1,a1); hold on;

grpdelay(b2,a2); legend('Butterworth','Chebyshev-I');

% Pole-Zero Plot figure;
subplot(121),zplane(b1,a1),title('Butterworth')
subplot(122),zplane(b2,a2),title('Chebyshev-I')

% Impulse Response figure;
subplot(211),impz(b1,a1),title('Butterworth Impulse')
subplot(212),impz(b2,a2),title('Chebyshev Impulse')

% Pole Magnitude figure;
subplot(121),stem(abs(roots(a1)),'filled') title('Butterworth Pole Magnitude')

subplot(122),stem(abs(roots(a2)),'filled') title('Chebyshev Pole Magnitude')

disp(['Butterworth Order = ',num2str(n1)]) disp(['Chebyshev Order = ',num2str(n2)])

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